

Genomic Risk Assessment and Early Intervention for Rare Genetic Disorders

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Abstract:

Genetic diseases pose significant challenges due to their complex nature and the need for specialized diagnostic techniques. This research focuses on developing a comprehensive framework for classifying genetic diseases using advanced machine learning algorithms. By applying techniques such as random forest classification, artificial neural networks, and support vector machines, the proposed framework aims to extract complex genetic data and accurately identify various genetic diseases. This approach empowers individuals to become more aware of their genetic health and take proactive steps for prevention and early intervention. The research contributes to the field of healthcare by providing a data-driven solution for genetic disease classification. By integrating new technologies like the chain classifier approach, the framework enhances diagnostic accuracy and addresses the complexities associated with genetic interactions. This work represents a significant step towards improving the understanding and management of genetic diseases, ultimately contributing to global health.

Keywords: Genetic Disorders, Hereditary, Random Forest, Artificial Neural Network (ANN), Support Vector Machine (SVM), Machine Learning, Chain Classifier Approach.

I. INTRODUCTION

Genetic disorders are caused by alterations in a person's genetic code (DNA), which lead to changes in an organism's function. Genomic data provides vital health indicators and is analyzed to better understand these genetic diseases. Genomics, a branch of bioinformatics, focuses on the study of genomes and their abnormalities [1]. Genetic disorders are classified into single-gene, mitochondrial, and multifactorial inheritance disorders, each examined based on DNA structure. Single-gene disorders result from mutations in a single gene, while mitochondrial disorders arise from deletions or replacements in mitochondrial DNA, a non-nuclear part of the genetic

structure. Multifactorial disorders are caused by a combination of environmental factors and mutations in multiple genes [2].

Genes in DNA encode information for various proteins, and any alteration can lead to abnormal protein formation, disrupting cellular function and causing genetic disorders. Out of the 7.8 billion people on Earth, it is estimated that 350 million—nearly 4.5%—are living with a rare genetic disorder. While this may seem like a small proportion, it is crucial to consider that 95% of these diseases are incurable [3]. Furthermore, 85% of individuals diagnosed with genetic disorders receive their diagnosis late, primarily due to a lack of awareness about these conditions. Genetic disorders are categorized into three main classes: mitochondrial, single-gene, and multifactorial genetic disorders [4]. To address the challenges of early diagnosis, we have developed a machine learning model capable of predicting the class of genetic disorder based on an individual's genomic data [5].

II. LITERATURE REVIEW

Collectively, these papers highlight the transformative role that machine learning can play in genetic disease prediction and genomic medicine. They showcase potential applications in various medical fields, including cancer, dementia, and diabetes. Furthermore, they emphasize the need for collaboration and data sharing to accelerate progress in genomic medicine, particularly to address the challenge of identifying disease-causing mutations. The

unique challenges and opportunities of rare genetic disease research in India are also explored, underscoring the contribution of genomics to the understanding of rare diseases through initiatives like the India Alliance Network [6]. The application of machine learning in genetics and genomics presents a promising avenue to revolutionize biomedical research [7]. One study proposes a novel classification method for predicting disease and genetic subtypes, demonstrating improved accuracy compared to traditional methods. Another study highlights the effectiveness of machine learning in identifying disease genes through the functional similarity of genes, assessed using Gene Ontology (GO) annotations. Together, these research works highlight the growing importance of incorporating machine learning into genomics to enhance our understanding of the genetic basis of disease. This integration is paving the way for personalized medicine and improved patient outcomes [5, 8].

III. DATASET DESCRIPTION

The large, reliable dataset used in this study was carefully curated from various sources to support accurate disease prediction and provide expert recommendations in healthcare settings. This dataset was compiled through extensive interviews with medical specialists, data collected from several hospitals, and a thorough analysis of the existing literature. It encompasses a broad spectrum of illnesses categorized across multiple medical disciplines, including detailed records of symptoms reported by patients of different ages, providing a comprehensive view of symptomatology across diverse demographic groups. Each record in the dataset was verified for accuracy by cross-referencing it with medical standards, and patient identities were anonymized to ensure privacy and compliance with ethical guidelines. This exhaustive collection of symptoms and associated disorders not only enhances the dataset's reliability but also underscores its value for developing clinical decision support systems and training machine learning models aimed at improving diagnostic precision and patient care outcomes.

IV. PROPOSED METHOD

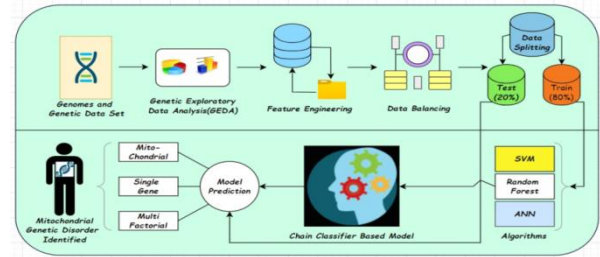


Figure 1: Architecture of the proposed work

Fig. 1 shows the actual flow of the work. The process starts with collection of the data and then this data is turned into visual form which helps in finding hidden patterns and then in the part of feature engineering the data is normalized and then balanced in order to avoid biased results [9]. The resultant data is split into parts 20% for testing and the rest for training which is used to train the model with the three algorithms (SVM, RANDOM FOREST, ANN) which forms the chain classifier model and the other part is used to test the model.

Data Pre-processing

Data pre-processing is a crucial phase in both data analysis and machine learning workflows, playing a critical role in refining raw data for optimal use. The process involves addressing missing values, removing duplicates, and rectifying errors during data cleaning. Transformative steps include normalizing numerical features and standardizing data by encoding categorical variables [10]. Techniques such as dimensionality reduction and feature selection are employed to reduce data complexity. Additionally, handling outliers and addressing class imbalances contribute to the robustness of model performance. Dividing the data into training, validation, and test sets, as well as accounting for temporal aspects in time series data, are integral parts of the pre-processing approach [9, 11]. The iterative nature of this process allows it to adapt to the unique characteristics of the dataset, ultimately enhancing data quality and suitability for specific analytical or modeling tasks.

Data Cleaning

In the dataset considered, numerous columns were deemed irrelevant for meaningful analysis, including patient name, family name, patient ID, institute location, institute name, father's name, and place of birth, as they did not provide insights related to patient health or hereditary factors. To address missing values, numeric gaps were filled with the median, while categorical variables were backfilled to maintain data continuity. Notably, the Genetic Disorder and Subclass Disorder columns, which served as the target variables, had missing values. To preserve data integrity, rows with missing values in these columns were excluded. The resultant dataset for the Genetic Disorder neural network comprised 65,190 data points and 46 columns, with Genetic Disorder as the target variable. Similarly, the dataset for the Subclass Disorder neural network included 65,190 data points and 47 columns, with Subclass Disorder as the target variable.

Exploratory Data Analysis

A correlation matrix shown in figure 2 is a statistical tool used in exploratory data analysis to quantify the strength and direction of relationships between numerical variables in a dataset. It consists of correlation coefficients ranging from -1 to 1, indicating negative to positive associations, with 0 representing no correlation. Visualizing the matrix through techniques like heatmaps provides a quick overview, helping identify significant correlations and patterns among variables.

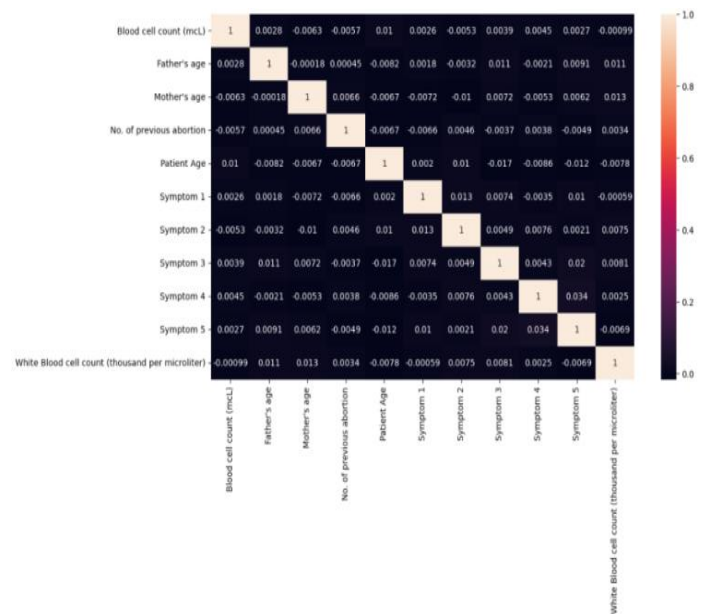


Figure 2: Correlation Matrix

Density Plot and Table of Results:

A density plot shown in figures 3 and 4 can effectively illustrate the distribution of data points for three distinct genetic disorders – mitochondrial, single-gene, and multifactorial. In this plot, the x-axis typically denotes a relevant variable such as age of onset or severity, while the y-axis represents the probability density [12]. The resulting curves showcase the concentration and distribution patterns of data points for each disorder, with peaks indicating areas of higher density. This visualization facilitates a comparative analysis, enabling a clear understanding of the unique characteristics and prevalence of each genetic disorder within the dataset.

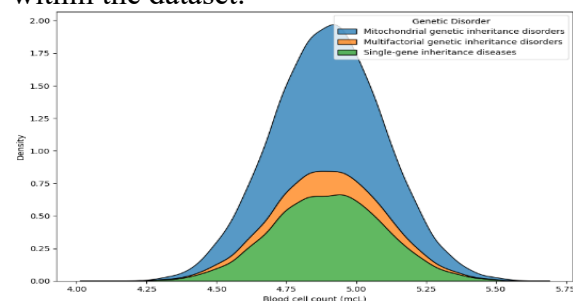


Figure 3: Density Plot for Blood Cell Count (mcL)

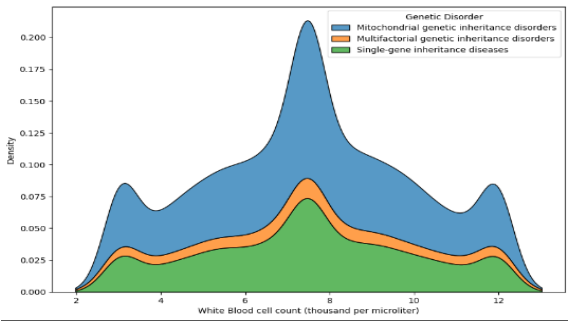


Figure 4: Density Plot for White Blood Cell Count (thousand per microliter)

Table 1: Density Plot Results for Blood Cell Count (mcL) grouped by ‘Genetic Disorder’

Genetic Disorder	Count	Mean	Std	Min	25%	50%	75%	Max
0	12348	7.47	2.514	3.0	5.666	7.477	9.279	12.0
1	2071	7.52	2.644	3.0	5.529	7.477	9.516	12.0
2	7664	7.48	2.498	3.0	5.666	7.477	9.238	12.0

Table 2: Density Plot Results for White Blood Cell Count (1000 per microliter)

Genetic Disorder	Count	Mean	Std	Min	25%	50%	75%	Max
0	12348	4.89	0.200	4.092	4.763	4.900	5.033	5.592
1	2071	4.89	0.199	4.248	4.762	4.892	5.032	5.515
2	7664	4.89	0.198	4.215	4.763	4.899	5.034	5.609

grouped by ‘Genetic Disorder’

- Box Plot

Figure 5 represents the box plots for "Blood cell count (mcL)" and "White Blood cell count (thousand per microliter)" against a specified target variable. Box plots, visually depicting the distribution and central tendency of numerical data, help identify potential outliers and reveal distributional characteristics. Each plot showcases the interquartile range, median, and potential outliers beyond whiskers, providing insights into the spread and skewness of blood cell counts concerning the target variable [11:13].

This visualization aids in discerning patterns and variations within the dataset associated with the specified blood cell attributes. Figure.5 provides key statistics. It also identifies outliers, helping to compare distributions across groups.

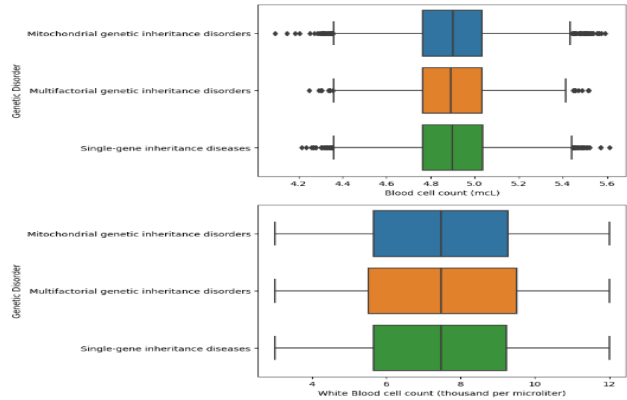


Figure 5: Box Plot for Blood Cell Count and White Blood Cell Count

Table 3: Box Plot Results for Blood Cell Count (mcL) grouped by ‘Genetic Disorder’

Genetic Disorder	Count	Mean	Std	Min	25%	50%	75%	Max
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- Count Plot

As shown in figure 6, Sea-born is used to create count plots for categorical data in the dataset. The initial count plot displays the

distribution of the target variable by counting occurrences of each category along with the x-axis. The second count plot introduces a comparison between the target variable and patient age, incorporating the hue parameter to distinguish categories within the target variable. These count plots are valuable for visually representing the frequency of each category, providing an overall view of the target variable's distribution and a more detailed insight into its relationship with patient age.

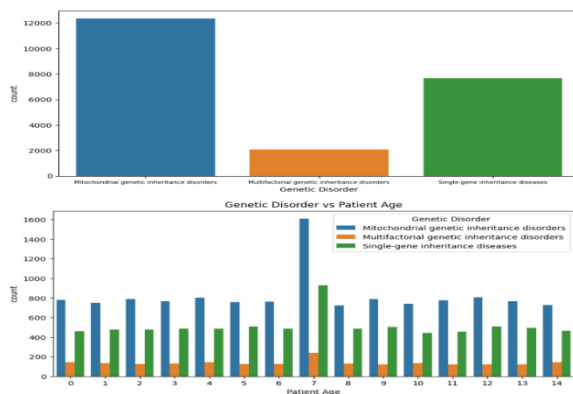


Figure 6: Count Plot

3.2.7 Algorithm

In the employed classification strategy, Support Vector Machines (SVM), Random Forest, and Artificial Neural Networks (ANN) were deliberately selected to cater to distinct dataset characteristics. SVM's adaptability is suited for scenarios with clear class margins, Random Forest's robustness and interpretability contribute to accuracy and comprehension, and ANN performs well in detecting complex patterns in datasets featuring non-linearities. The algorithmic choices were aimed at balancing the utilization of each model's strengths. Furthermore, a sequential learning approach was introduced, integrating SVM, Random Forest, and ANN. This process involved using the output predictions from one model as supplementary input features for the subsequent model, harnessing the individual strengths of each algorithm. This ensemble method proves particularly advantageous in situations involving complex relationships and diverse patterns, ultimately augmenting the precision and resilience of the

comprehensive classification system and the steps involved are [14, 15].

Step 1: Data Collection

Collect genetic data to serve as input for the chain classification model.

Step 2: Labelling Genetic Dataset

Assign labels to the genetic dataset, specifying the targeted sections or classes.

Step 3: Data Pre-processing

Perform data pre-processing, including tasks like handling missing values and normalizing features, to enhance data quality.

Step 4: Data Splitting

Divide the genetic dataset into training and testing sets for model evaluation.

Step 5: Model Training and Validation

Train the chain classification model using algorithms such as SVM, Random Forest, and ANN, implementing a chain classification approach.

Step 6: Chain Classification Implementation

Execute the chain classification approach by utilizing output predictions from one model as input features for the subsequent model.

Step 7: Model Validation

Validate the model's performance using the testing dataset.

Step 8: Accuracy Evaluation

Assess the accuracy measures to gauge the effectiveness of the chain classification model.

Step 9: Fine-Tuning and Optimization

Fine-tune model parameters and optimize algorithmic choices based on accuracy measures.

Step 10: Results Analysis and Interpretation.

VI. Experimental Analysis

• Dataset

The Genetic Variant Classification dataset on Kaggle is a collection of over 650,000 genetic variants and their corresponding clinical classifications, created by the ClinVar consortium. It includes information on the variant's location on the genome, type (e.g., single nucleotide polymorphism, deletion),

and clinical significance (e.g., benign, pathogenic). The dataset is used by researchers and clinicians to study the genetics of human disease, identify new disease-causing variants, develop diagnostic tests, and inform genetic counselling. The dataset has a dataset size of 65188 records and is formatted in CSV. It includes features such as chromosome, position on the chromosome, reference allele, alternate allele, clinical classification, variant type [7], significance sign, sequence ontology term, gene symbol, depth, allele frequency, genotype, filter status, and additional information about the variant.

The Genomes and Genetics database on Bio-Portal is a vast collection of ontologies and resources related to genomics and genetics. It has 31897 rows and 45 columns and offers a central location for researchers and clinicians to access and utilize this information to advance their understanding of the genetic basis of human disease. The database contains over 100 ontologies, including Gene Ontology (GO), Human Gene Mutation Database (HGMD), Online Mendelian Inheritance in Man (OMIM), Clinical Genome Interpretation (CGI) Ontology, and Disease Ontology (DO). The database features a search and browse interface, annotations and links to external resources, and a download option for users to use in their research and applications [9].

• Results and Discussion

Table 5 shows precision, recall, F1 score, and support for Class 1 and Class 2.

Class 1: With a precision of 0.92, recall of 0.93, and F1 score of 0.92, the model is highly effective at identifying Class 1 instances.

Class 2: The precision is 0.89, recall is 0.90, and F1 score is 0.89, indicating reliable performance for Class 2.

Overall Accuracy: The model achieves an overall accuracy of 0.91, demonstrating its

effectiveness in identifying genetic disease risks.

Support: The dataset contains 2470 samples for Class 1 and 1533 for Class 2, showing a class imbalance.

Average Metrics: The average precision, recall, and F1 score are 0.91, reflecting strong and consistent model performance.

Implications: The results suggest the model is accurate and resilient, suitable for identifying genetic disease risks across different data sets.

The accuracy curves shown in figure 7 indicate that both training and validation accuracy quickly rise and then stabilize around the 10th epoch. The similar trends and stability of both curves suggest the model generalizes well and has converged without overfitting.

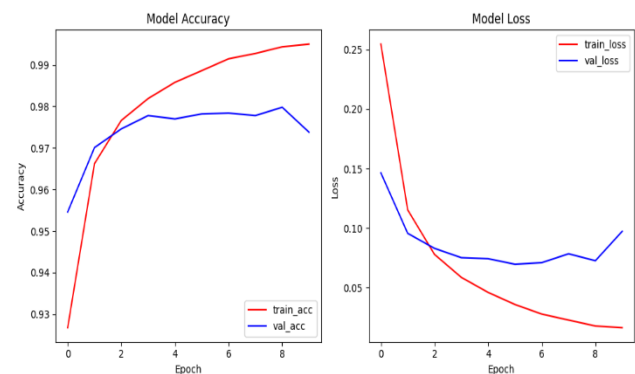


Figure 7: Loss and Accuracy curves of ANN model

Table 5: Classification Report of Chain Classifier Based Model

Algorithm	Accur acy	Precisi on	Reca ll	F1- Score	Supp ort
Support Vector Machine	0.90	0.91	0.89	0.90	1553
Random Forest	0.92	0.91	0.89	0.90	1533
Artificial Neural Networks	0.93	0.92	0.89	0.88	1531
Chain Classifier Model	94	0.91	0.90	0.90	1551

IX. CONCLUSION

This research presents a novel approach to identifying genetic diseases using a combination of Support Vector Machine, Random Forest, and Artificial Neural Networks. By carefully comparing the performance of these methods, we have identified the most effective algorithm for each specific disease. The use of balancing methods ensures that our test scores are fair and accurate, even for less common diseases. The provided graphs offer a clear visualization of the performance of each algorithm, facilitating informed decision-making. By combining these three algorithms, we have significantly improved the accuracy of predictions, leading to more reliable and effective diagnosis of genetic diseases. This research contributes to the advancement of genetic disease diagnosis and has the potential to improve patient outcomes.

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